
ASSIGNMENT

PROGRAMMING USING C

WEEK – 01

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Week – 1(a)

We're starting out by printing the most famous computing phrase of all time! In the editor below, use either printf or cout to print the string **Hello, World!** to stdout.

Input Format

You do not need to read any input in this challenge.

Output Format

Print **Hello, World!** to stdout.

```
1 #include<stdio.h>
2 int main()
3 {
4     printf("Hello, World!");
5
6 }
7
```

	Expected	Got	
✓	Hello, World!	Hello, World!	✓

Passed all tests! ✓

This challenge will help you to learn how to take a character, a string and a sentence as input in C.

To take a single character **ch** as input, you can use `scanf("%c", &ch);` and `printf("%c", ch)` writes a character specified by the argument `ch` to stdout:

```
char ch;  
scanf("%c", &ch);  
printf("%c", ch);
```

This piece of code prints the character **ch**.

Task

You have to print the character, **ch**.

Input Format

Take a character, **ch** as input.

Output Format

Print the character, **ch**.

```
1  #include<stdio.h>  
2  int main()  
3  {  
4      char ch;  
5      scanf("%c",&ch);  
6      printf("%c",ch);  
7  }
```

	Input	Expected	Got	
✓	C	C	C	✓

Passed all tests! ✓

Your task is to take two numbers of `int` data type, two numbers of `float` data type as input and output their sum:

1. Declare **4** variables: two of type `int` and two of type `float`.
2. Read **2** lines of input from `stdin` (according to the sequence given in the 'Input Format' section below) and initialize your **4** variables.
3. Use the `+` and `-` operator to perform the following operations:
 - o Print the sum and difference of two `int` variable on a new line.
 - o Print the sum and difference of two `float` variable rounded to one decimal place on a new line.

Input Format

The first line contains two integers.

The second line contains two floating point numbers.

Constraints

- $1 \leq \text{integer variables} \leq 10^4$
- $1 \leq \text{float variables} \leq 10^4$

```
1  #include<stdio.h>
2  int main()
3  {
4      int a,b;
5      float c,d;
6      scanf("%d %d %f%f",&a,&b,&c,&d);
7      printf("%d %d \n",a+b,a-b);
8      printf("%.1f %.1f \n",c+d,c-d);
9
10 }
```

	Input	Expected	Got	
✓	10 4 4.0 2.0	14 6 6.0 2.0	14 6 6.0 2.0	✓
✓	20 8 8.0 4.0	28 12 12.0 4.0	28 12 12.0 4.0	✓

Passed all tests! ✓

Week – 1(b)

The code given below contains instructions to print the text "**I love Apples**" to the console.

The `\n` in the text "**I love Apples\n**" ensures that the line breaks after printing the text "**I love Apples**" (which means that nothing else is printed on the same line).

Follow the steps given below to change the text, execute **compile** command and finally **execute** the file :

1. In the code given below, change the text to print "**I love Mangoes**" instead of "**I love Apples**".

```
#include <stdio.h>

int main()
{
    printf("I love Mangoes");
    return 0;
}
```

	Expected	Got	
✓	I love Mangoes	I love Mangoes	✓

Passed all tests! ✓

Given below is a simple program written in **C** language.

Change the text in the code given below to make the program print "**Hello C**" instead of "**Hello B**".

```
#include <stdio.h>

int main()
{
    printf("Hello C");
    return 0;
}
```

	Expected	Got	
✓	Hello C	Hello C	✓

Passed all tests! ✓

Week – 1(c)

Write a program to input a name (as a single character) and marks of three tests as m1, m2, and m3 of a student considering all the three marks have been given in integer format.

Now, you need to calculate the average of the given marks and print it along with the name as mentioned in the output format section.

All the test marks are in integers and hence calculate the average in integer as well. That is, you need to print the integer part of the average only and neglect the decimal part.

```
#include<stdio.h>
int main()
{
    char name;
    int a,b,c,avg;
    scanf("%c",&name);
    scanf("%d%d%d",&a,&b,&c);
    avg=(a+b+c)/3;
    printf("%c\n",name);
    printf("%d",avg);
}
```

	Input	Expected	Got	
✓	A 3 4 6	A 4	A 4	✓
✓	T 7 3 8	T 6	T 6	✓
✓	R 0 100 99	R 66	R 66	✓

Passed all tests! ✓

Some C data types, their format specifiers, and their most common bit widths are as follows:

- *Int* ("%d"): 32 Bit integer
- *Long* ("%ld"): 64 bit integer
- *Char* ("%c"): Character type
- *Float* ("%f"): 32 bit real value
- *Double* ("%lf"): 64 bit real value

```
#include<stdio.h>
int main()
{
    int a;
    long b;
    char c;
    float d;
    double e;
    scanf(" %d %ld %c %f %lf",&a,&b,&c,&d,&e);
    printf("%d \n%ld \n%c \n%0.3f \n%0.9lf",a,b,c,d,e);
}
```

	Input	Expected	Got	
✓	3 12345678912345 a 334.23 14049.30493	3 12345678912345 a 334.230 14049.304930000	3 12345678912345 a 334.230 14049.304930000	✓

Passed all tests! ✓

Write a program to print the [ASCII value](#) and the two adjacent characters of the given character.

Input

E

```
#include<stdio.h>
int main()
{
    char a;
    scanf("%c",&a);
    printf("%d\n",a);
    printf("%c %c",a-1,a+1);
}
```

	Input	Expected	Got	
✓	E	69 D F	69 D F	✓

Passed all tests! ✓